

Network problem	Example	Classification
Faulty network layout	Multiple cascading of switches on fast networks.	Hardware
Faulty network cabling	Cable sequence not observed while crimping; faulty crimping.	Hardware
Faulty device on the network	A faulty Ethernet card transmitting a lot of packets.	Hardware
Various network protocols installed on the same network causing network performance degradation	Unwanted protocols like NetBEUI, IPX installed on IP networks.	Network configuration
Malware present on the network	Various back doors like Back Orifice	Network

Table 1. Possible reasons for network related problems

different devices present on the network. You may have different operating systems with various protocols, network printers, wireless access points, firewalls, routers, managed switches, etc.

Troubleshooting should start with monitoring network traffic continuously over a period of time. This monitoring should cover various parameters, including the protocols being used, bandwidth utilisation, activity graphs, traffic direction and flow. The output from this monitoring should then be analysed for possible anomalies, to pinpoint the problem. A good, reliable network monitoring tool becomes a must to pinpoint such issues.

There are various open source networking tools available on the Linux platform, including packet sniffers like Wireshark, Kismet and Tcpcdump, and network probes like Hping and Traceroute. However, the moment you consider simple network monitoring, the automatic choice should be Ntop!

It is said that the good things in life are never free, but I beg to differ. Ntop, developed by Luca Deri, an Italian scientist, is available free of cost under a GNU licence. It is really a very effective way to monitor the network, and has various functions from packet capture to displaying an analysis of the captures in a Web browser, in different modules:

- **Packet capture:** Ntop uses Linux-based *libpcap* to capture packets on the network. Typically, these libraries have a low buffer capacity. Ntop captures packets from the predefined network interface and buffers them, reducing the possibility of packet drops.
- **Packet analyser:** Ntop analyses the captured packets one raw packet at a time, to extract information including the host's operating system, a record of when the host appeared on the network for the first time, the protocol, address resolution, etc. The information is cached and stored in an SQL database with a user-definable path.
- **Network flow measurement:** Network flow can be defined as matching captured packet(s) with a predetermined rule or criterion, and incrementing a flow counter by one, whenever a captured packet matches the predefined rule.
- **Plug-ins:** Various plug-ins (dynamically loaded libraries) can be enabled or disabled at start-up, or from the Web interface.
- **Web server:** All the captured, analysed and measured information is displayed on a Web page for easy navigation, inspection and analysis by the user. By default, the Ntop Web server listens on TCP port 3000

for HTTP connections; HTTPS is also supported. The type of connection and the port numbers to listen on can be specified via command-line options. Authentication is required to access data in both cases.

Installing Ntop in Linux is simple and straightforward. Under Ubuntu, it is a simple command line:

```
sudo apt-get install ntop
```

For SuSE enthusiasts, the best way to check repositories for the latest version of Ntop is to search <http://www.packages.opensuse-community.org>. To install Ntop, just click the *Install* button! At the time of writing this article, repositories were available for OpenSuSE 11.x for Ntop versions 3.3 and 3.3.10.

Monitoring traffic using Ntop

To start, run Ntop from the command prompt, with the required parameters—for example:

```
ntop -i eth0 -w 1989 -W 2988 -A-Pwd123
```

- With this command line, Ntop will:
- capture packets from the *eth0* interface
- run the HTTP Web server on port 1000
- run the HTTPS Web server on port 2000
- assign *Pwd123* as the Ntop administrator password for the Web interfaces

- Ntop has some other interesting command line parameters, including:
- *-p* limits Ntop to monitoring only the specified protocols.
 - *-P* can define directories used to store Ntop databases. Ideally, only the Ntop user should have access to this directory.
 - *-f TrafficDumpFile* specifies a file output from any packet-capture software like Wireshark. This may be used for offline analysis of the traffic. Invoking this option will stop captures from network cards.
 - *-d* will run Ntop in daemon mode (no output on the screen)
 - *-L* will log *stdout* to the *syslog* facility (useful along with *-d*)

Real-life examples

Networking engineers face challenging network congestion problems on various sites. It would have been a very difficult